

Compound interest tables



- 1 \$4400 is invested for three years at a rate of 5% per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	—	\$4400	\$220
Second year	$\$4400 + \220	\$4620	\$231
Third year	$\$4620 + \$\square$	$\$ \square$	$\$ \square$
Fourth year	$\$4851 + \$\square$	$\$ \square$	—

b Calculate the total interest earned over the three years.

- 2 \$3600 is invested for three years at a rate of 7% per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	—	\$3600	\$252
Second year	$\$3600 + \252	\$3852	\$269.64
Third year	$\$3852 + \$\square$	$\$ \square$	$\$ \square$
Fourth year	$\$4121.64 + \$\square$	$\$ \square$	—

b Calculate the total interest earned over the three years.

- 3 \$570 is invested for two years at a rate of 9% per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	—	\$570	$\$ \square$
Second year	$\$570 + \$\square$	$\$ \square$	$\$ \square$
Third year	$\$ \square + \55.92	$\$ \square$	—

b Calculate the total interest earned over the three years.

- 4 \$6100 is invested for two years at a rate of  per year, compounded annually.

a Copy and complete the table to determine the final value of the investment.

	Balance + interest	Total balance	Interest earned
First year	—	\$6100	\$1
Second year	\$6100 + \$1	\$1	\$1
Third year	\$1 + \$387.96	\$1	—

b Calculate the total interest earned over the three years.

- 5 \$1000 is invested at 2% per year, compounded annually. The table tracks the growth of the principal over three years.

Time Period n (years)	Value at beginning of time period	Value at end of time period	Interest earned in time period
1	\$1000	\$1	\$1
2	\$1	\$1040.40	\$1
3	\$1040.40	\$1061.21	\$1

a Copy and complete the table.

b Calculate the total interest earned over the three years.

Compound interest

- 6 State whether the following statements are true or false about compound interest:

- a Interest is earned on the principal.
- b The interest in any time period is calculated using only the original principal.
- c Interest is earned on any accumulated interest.
- d The amount of interest earned in any time period changes from one period to the next.

- 7 Roberto's investment of \$70 000 earns interest at 5% per year, compounded annually over 3 years.

- a Calculate the value of the investment after 3 years.
- b Calculate the amount of interest earned.

- 8 Maria invested \$8090 at 8% per year, compounded annually over 3 years. Answer the following questions without using the compound interest formula.
- What was the interest for the first year?
 - What was the total amount invested after the first year?
 - What was the interest for the second year?
 - What was the total amount invested after the second year?
 - What was the interest for the third year?
 - What was the total amount invested after the third year?
 - What was the total amount of interest earned over the 3 years?
 - What was the interest as a percentage of the initial investment? Give your answer as a percentage to two decimal places.
 - What would have been the simple interest earned after 3 years?
 - How much more compound interest than simple interest would have accumulated over these 3 years?
- 9 Hannah borrows \$5000 at a rate of 4.2% per year, compounded annually.
- After 4 years, Hannah repays the loan all at once. How much money does she pay back in total?
 - How much interest was generated on the loan over the four years?
- 10 Jerome has just won \$50 000. When he retires in 15 years, he wants to have \$94 000 in his fund which earns 6% interest annually. Determine how much of his winnings he needs to invest now to achieve this.
- 11 Victoria has been promised an inheritance of \$70 000 in 5 years time. Determine the maximum she can borrow now at a rate of 7% compounded annually, and still be able to pay off the loan with her inheritance.
- 12 When Yuri got his first job, he invested \$10 000 in an account with a 1.3% annual growth rate.
- Write a function, $A(t)$, that represents the value of this investment t years after Yuri started his job.
 - Calculate how much more the investment will be worth when Yuri has been working for 12 years than when he had been working for 5 years.



Other compound periods

13 Calculate the value of the following investments:

- a** A \$7230 investment earns interest at 3% per year, compounded quarterly over 7 years.
- b** A \$5520 investment earns interest at 0.8% per year, compounded monthly over 2 years.
- c** A \$1980 investment earns interest at 0.45% per year, compounded weekly over 2 years.
- d** A \$3000 investment earns interest at 4.2% per year, compounded daily over 14 years.

14 Calculate the amount of interest earned by the following investments:

- a** Beth's investment of \$4800 earns interest at 3.3% per year, compounded monthly over 13 years.
- b** James's investment of \$4210 earns interest at 0.78% per year, compounded weekly over 16 years.
- c** Davinda's investment of \$5390 earns interest at 2.1% per year, compounded daily over 11 years.
- d** Una's investment of \$80 000 earns interest at 0.5% per year, compounded semi-annually over 10 years.

15 Amelia's investment into a 15-year 4.09% per year. corporate bond grew to \$14 160. Calculate the size of Amelia's initial investment if interest was compounded:

- a** Annually **b** Semiannually **c** Quarterly **d** Monthly
- e** Weekly **f** Daily

16 Grace wants to invest \$1600 at 5% per year for 7 years. She has three investment options, compounded quarterly, compounded monthly, or compounded weekly.

- a** Calculate the value of the investment if it is compounded:
 - i** quarterly **ii** monthly **iii** weekly
- b** Find the difference between the final investment values of the three options.
- c** Describe the effect that changing the compound frequency has on the interest earned.

17 Maria has \$5000 to invest for 6 years and would like to know which investment plan to enter into out of the following three:

- Plan 1: invest at 4.70% per year interest, compounded monthly
- Plan 2: invest at 5.77% per year interest, compounded quarterly
- Plan 3: invest at 5.40% per year interest, compounded annually

Determine which is the best option for Maria. Use calculations and reasoning to explain your

